Verification Report

Assessment of the NuLife GreenTech: Ontario Avenue Bio-Oil Sequestration Project

Version Control

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Report for:

Isometric and NuLife

https://isometric.com/
https://www.nulifegreentech.com/

Assured by:

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Formal verification of project

Normec Verifavia has been contracted by Isometric on behalf of NuLife to conduct a verification of the NuLife bio-oil carbon removal project against the requirements specified in the Isometric Bio-Oil Geological Storage Protocol (V1.1) and associated modules, including the Feedstock Characterization, Energy Use Accounting, Transportation Accounting, and Storage Accounting modules. The objective of the verification is to confirm that the carbon removals claimed by NuLife for the reporting period from September 6th to November 3rd are within scope, real, quantifiable, additional, verifiable, counted once, under clear ownership, and compliant with all applicable monitoring and reporting requirements.

Normec Verifavia conducted this verification to provide assurance to Isometric that the GHG Statement, life-cycle analysis (LCA) inputs, operational data, sampling records, and reporting practices for the period under review are free from material misstatement and non-conformance. Following the receipt of NuLife's submission package, Normec Verifavia performed a risk assessment to determine the sampling and verification methodology appropriate for this reporting period.

The verification team reviewed supporting documentation, including feedstock receipts, operational logs, bio-oil sampling results, energy use records, transport documentation, and injection site data, in accordance with the approved verification plan and in alignment with the applicable verification standards (ISO 14064-3, ISO/IEC 17029 and ISO 14065)

1. Introduction

1.1 Project Presentation

NuLife GreenTech Inc. is a Canadian carbon removal company based in Saskatoon, Saskatchewan. The company develops and operates a facility that use Hydrothermal Liquefaction (HTL) technology to convert waste biomass into a stable, carbon-rich bio-oil. Instead of being landfilled or left to decompose, waste streams such as dairy processing sludge, oat residues, agricultural by-products, and fats, oils and greases are diverted and transformed into bio-oil.

This bio-oil is then permanently stored in deep underground salt caverns through a regulated injection process. These caverns, formed millions of years ago and historically used for natural gas and liquid fuel storage, provide a secure geological environment where carbon is locked away for more than 1,000 years.

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By combining waste management with durable carbon storage, NuLife creates a net-negative climate solution:

- Preventing emissions from waste decomposition or land application,
- Transforming biomass carbon into a stable form, and
- Permanently isolating that carbon from the atmosphere.

In line with the requirements in Isometric's Standard, Protocol and relevant Modules, NuLife quantifies, monitors, and verifies the carbon removed by its operations. The resulting high-quality Carbon Dioxide Removal (CDR) credits help organizations and governments meet their climate commitments while accelerating the development of innovative carbon removal infrastructure in Canada.

1.2 Scope

Normec Verifavia has been appointed by Isometric to provide independent assurance on the quality of data and methodology used to assess performance against key indicators under the Isometric standard, including:

- Measurement and reporting of carbon removal volumes associated with the Hydrothermal Liquefaction (HTL) process and subsequent bio-oil storage.
- Monitoring of biomass feedstock sourcing and treatment volumes at the Saskatoon facility.
- Verification of data supporting NuLife's capacity expansion and operational scaling.

To confirm this information and its coherence with Isometric's requirements, an on-site visit was conducted on the 20th of August 2025.

1.3 Audit Methodology

The assurance process included interviews with NuLife staff responsible for monitoring and measuring plant performance to gain insight into the data collection process and to evaluate the quality of the data used for reporting. In addition, the audit team conducted both on-site validation and verification activities at the Saskatoon facility and desk-based reviews of supporting documentation, records, and calculations.

The audit methodology was designed with the following key objectives:

- To confirm that project activities and monitoring practices align with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1).
- To assess whether carbon removal volumes are real, quantifiable, additional, and verifiable.
- To determine whether reported data is free from material misstatement and non-conformance.

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- To evaluate that monitoring and reporting systems provide consistent, transparent, and traceable records.
- A validation plan was shared with NuLife in advance (see Appendix 3), outlining the criteria, scope, and sampling strategy. This plan specified which data streams, calculations, and operational records would be reviewed through interviews and site inspection, and which would be validated through desk review.
- Site visit activities included: observation of HTL processing operations, review of sampling and testing procedures, confirmation of feedstock receiving and weighing processes, inspection of storage tanks, and examination of safety and monitoring protocols.
- Desk review activities included: assessment of life-cycle analysis (LCA) models, validation of emission and removal calculations, review of Renewable Energy Certificate (REC) purchases, confirmation of biomass sourcing documentation, transport and injection records, and brine quality monitoring data.
- Risks identified in the validation methodology—such as reliance on third-party laboratories, ownership of carbon attributes, and uncertainty in feedstock carbon content—were mitigated by adopting conservative assumptions, reviewing multiple data sources, and requiring supporting documentation (e.g., supplier attestations, injection site agreements, calibration certificates).
- Normec Verifavia conducted the independent assurance audit based on the following criteria:
    - ISO/IEC 17029:2019 – Conformity assessment – General principles and requirements for validation and verification bodies.
    - ISO 14065:2020 – General principles and requirements for bodies validating and verifying environmental information.
- The GHG Protocol – Corporate Accounting and Reporting Standard.
- Normec Verifavia AMSM (Version 9.9, July 2025).
- Isometric Standard v1.7.3 (July 2025).
- Isometric Bio-Oil Geological Storage Protocol (V1.1).
- Isometric Modules:
    - Biomass Feedstock Accounting (V1.2)
    - Bio-oil Storage in Salt Caverns (V1.1)
    - Energy Use Accounting (V1.2)
    - Transportation Emissions Accounting (V1.1)
    - Embodied Emissions Accounting Module (V1.0)

1.4 Responsibilities

NuLife is solely responsible for preparing and reporting its carbon related data, including tracking files, calculation models, primary data, and any supporting information or assessments. Additionally, NuLife is responsible for defining, establishing, and maintaining

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appropriate performance management and internal control systems from which the reported information is derived.

In accordance with the assurance agreement, our responsibility is to form an independent opinion based on an examination of the information and data presented in the NuLife reporting system, Isometric's submissions, and other supporting documents, and to report that opinion to NuLife. Additionally, we report if, in our opinion:

- The presented data contains or may contain misstatements (omissions, misrepresentations, or errors) or non-conformities.
- The verification team has not received all necessary information and explanations required to conduct its examination.
- Improvements can be made to NuLife's performance in monitoring and reporting.

2. Technology, Processes, and Inputs

2.1 Baseline scenario

NuLife's documentation describes the baseline scenario as the *business-as-usual management of waste biomass in the Saskatoon region*. In the absence of the project, the identified feedstocks (e.g., dairy DAF sludge, oat hulls, from [redacted] bran, agricultural residues, fats, oils and greases) would typically be disposed of through land application, landfilling, or wastewater treatment. Under these counterfactual pathways, the carbon stored in the biomass would be released back into the atmosphere as biogenic CO_2 or methane within a relatively short timeframe.

Normec Verifavia validated this baseline description through:

- Review of feedstock supplier records and attestations confirming that the materials are treated as waste and have no viable market use as commodities.
- Cross-checking disposal practices in Saskatchewan for similar feedstocks to confirm that landfill, wastewater treatment, or land application is the standard management method.
- Assessment of counterfactual analysis in NuLife's PDD, which excludes any ineligible biomass (i.e., feedstocks that would have remained durably stored or could reasonably be used for energy generation).
- Interviews with NuLife staff and feedstock providers, which confirmed that the project does not divert biomass from existing long-term storage or energy applications.

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Based on this review, the baseline scenario was found to be consistent with protocol requirements: the waste biomass would otherwise decompose or be processed in a way that rapidly returns carbon to the atmosphere, and no durable storage of carbon would occur.

2.2 Project boundaries

NuLife's documentation defines the project boundaries to include all activities directly related to the conversion of biomass into bio-oil and its subsequent permanent storage. Specifically, the boundaries cover:

- Upstream feedstock handling – delivery of waste biomass to the Saskatoon HTL facility, including transport from local suppliers such as [redacted].
- HTL processing operations – conversion of biomass into bio-oil, wastewater, and tailgas at the Saskatoon plant.
- Onsite energy use and consumables – electricity (tracked by facility power meters, with RECs purchased to offset grid emissions), water use, and cleaning chemicals.
- Transport of bio-oil – road shipment by third-party haulers from Saskatoon to [redacted] storage site near [redacted].
- Geological storage – injection of bio-oil into regulated salt caverns, brine displacement and management via a disposal well, and required sampling and monitoring.

Normec Verifavia validated these boundaries through:

- Document review of NuLife's PDD, their submissions to Isometric, Isometric submissions, and supporting diagrams that delineate included and excluded activities.
- Site visit confirmation at the Saskatoon HTL plant, where feedstock receiving, storage, processing, and monitoring systems were observed.
- Inspection of transport documentation (e.g., bills of lading, delivery records) for bio-oil shipments to [redacted].
- Review of regulatory permits for [redacted] caverns and brine disposal well to confirm that injection and storage activities fall within project scope.

Based on this assessment, Normec Verifavia concluded that NuLife's defined project boundaries are consistent with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and cover all material sources and sinks relevant to carbon accounting.

2.3 Technology Overview

NuLife GreenTech operates a Hydrothermal Liquefaction (HTL) facility in Saskatoon, Saskatchewan, Canada. HTL is a continuous thermo-chemical process that converts high-moisture organic waste streams into a carbon-rich liquid bio-oil. This bio-oil is then transported to a licensed underground salt cavern near [redacted], where it is permanently sequestered in accordance with the Isometric Bio-Oil Geological Storage in Salt Caverns.

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The facility began design in 2022, installed its first processor in 2023, expanded with a second processor in 2024, and is scaling toward a 12-processor commercial module in Q4 2025. The plant is expected to operate through at least 2050.

2.4 Process Flow

1. **Feedstock Sourcing & Delivery**
    a. NuLife sources organic waste biomass within approximately a 25 km radius of Saskatoon.
    b. NuLife's feedstock providers are [redacted].
    c. Eligible feedstocks include dairy Dissolved Air Flotation (DAF) sludge, grain milling residues, oat hulls, agricultural residues, Fats, Oils and Greases (FOG), and other food manufacturing wastes.
    d. Dry biomass is stored onsite; wet biomass is held in 10,000 L storage tanks.
2. **HTL Conversion**
    a. Feedstock is mixed into slurry and pumped into the HTL processor.
    b. The process reforms the carbon in the waste into long-chain hydrocarbons (bio-oil).
    c. By-products include:
        i. Bio-oil – captured for storage.
        ii. Wastewater – recycled into the process or treated/disposed offsite.
        iii. Tailgas – ~99% biogenic CO_2, vented to atmosphere.
        iv. Biochar – emulsified with bio-oil and captured for storage when produced.
3. **Storage & Transport**
    a. Bio-oil is accumulated in external tanks until sufficient volume is available.
    b. A third-party hauler transports the bio-oil to the [redacted] injection site.
4. **Geological Injection**
    a. At the injection site, bio-oil is offloaded, pumped underground, and stored in a salt cavern.
    b. As bio-oil is injected, brine is displaced, monitored, and safely managed according to provincial permitting requirements.
5. **Human capital**
    a. NuLife Greentech has a total of 8 employees. Two of the employees' commute were taken into account for the life-cycle analysis (LCA).

2.5 Inputs and Supporting Materials

- **Biomass Feedstocks:** Organic wastes from food and agricultural sectors (DAF sludge, bran, oat hulls, milling residues, etc.).
- **Energy:** Primarily electricity from the Saskatchewan grid; Renewable Energy Certificates (RECs) are purchased to offset grid intensity.

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- **Consumables:** Small amounts of sodium hydroxide and hydrogen peroxide (for cleaning) activated carbon (for gas scrubbing), and municipal water for start-up/cooldown.
- **Transport:** Trucks are used for feedstock delivery, bio-oil transport, wastewater hauling, and sample delivery.

2.6 Conversion of biomass into bio-oil

The project achieves durable carbon removal by converting waste biomass into bio-oil and permanently storing it underground. Counterfactual analysis confirms that, without this project, the waste biomass would decompose, releasing its carbon back to the atmosphere.

On average, approximately 70% of the incoming biomass feedstock is converted into bio-oil through the Hydrothermal Liquefaction (HTL) process. Of this bio-oil, about 75% of the mass consists of elemental carbon. To calculate the amount of carbon dioxide removed and stored, NuLife applies the following formula:

$$1\ \text{MT}\ CO_2\ \text{sequestered} = 1\ \text{MT of bio-oil} \times 0.75(\text{carbon fraction}) \times 3.67(\text{molecular weight conversion of C to } CO_2)$$

This calculation reflects the stoichiometric conversion of carbon mass to its equivalent carbon dioxide mass, providing the basis for quantifying durable carbon removals under the Isometric protocol.

2.7 Carbon Removal Storage

NuLife quantifies stored carbon based on the mass of bio-oil produced, its measured carbon content, and the stoichiometric conversion of carbon to carbon dioxide. On average, approximately 70% of the incoming biomass feedstock is converted into bio-oil through the HTL process. the conversion of biomass to bio-oil is quantified on a daily basis via continuous mass and energy balance work, however, laboratory analysis shows that roughly 75% of the bio-oil mass consists of elemental carbon. Using the molecular weight conversion factor of 3.67 ($C \rightarrow CO_2$), NuLife applies the following calculation:

Total tCO_2e stored = bio-oil (kg) × fraction of elemental carbon content × stoichiometric conversion

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This verification report includes GHG accounting for four periods between 6th of September to 3rd of November.

Summary table of Total tCO_2e stored for GHG statement ID ggs_1K9D4YMZK1S01S57

Removal ID	*Removal period*	*tCO2e stored*
rmv_1K985KQ6Z1S0VVVA	23 Oct 2025 to 30 Oct 2025	59.55 tCO_2e
rmv_1K8VC1CFW1S02R3J	20 Oct 2025 to 22 Oct 2025	67.19 tCO_2e
rmv_1K8TY8T9P1S0A9MW	26 Sep 2025 to 19 Oct 2025	55.11 tCO_2e
rmv_1K6X92AJE1S0W10A	04 Sep 2025 to 25 Sep 2025	58.34 tCO_2e

This calculation approach aligns directly with the Isometric Bio-Oil Geological Storage Protocol (V1.1), which requires quantification of carbon removals based on measured carbon mass fractions and conversion to CO_2 equivalents. Bio-oil samples are collected three times per day and tested in duplicate onsite for moisture content. They are also sent to the University of Saskatchewan for independent elemental carbon analysis, including duplicate carbon wt% runs of each sample.

Once produced, the bio-oil is accumulated in external holding tanks and then transported by licensed third-party haulers to [redacted] injection site near [redacted]. At the site, the bio-oil is offloaded at an injection pad and pumped underground into deep salt caverns within the Prairie Evaporite Formation. Displaced brine is transferred to a regulated saltwater disposal well in the Lodgepole formation, with weekly onsite sampling and monthly laboratory tests confirming water quality.

Currently, two caverns (Caverns 1 and 4) are active with ~50,000 m^3 remaining each, while Cavern 5 (175,000 m^3) is permitted and expected to enter use once operational. The caverns were originally constructed for hydrocarbon storage in the 1950s–60s, are naturally pressurized at ~300 psi, and are subject to mechanical integrity testing every five years. No negative pressure events have been reported.

By combining accurate carbon accounting with regulated and geologically secure storage, NuLife ensures that the carbon embedded in its bio-oil is permanently isolated from the atmosphere for more than 1,000 years, in line with Isometric's permanence requirements.

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3. Verification Summary

3.1 Opinion

The verification confirms that NuLife's Saskatoon project uses a Hydrothermal Liquefaction (HTL) system to convert waste biomass into bio-oil for permanent geological storage in regulated salt caverns operated by [redacted].

The processes, inputs, monitoring activities, and reporting practices observed during the verification are consistent with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1). Operational controls, including feedstock input verification, daily bio-oil sampling, brine testing, equipment monitoring, and data recording, are implemented. Safety procedures, including mandatory visitor consent forms before entering the facility, are also in place.

3.2 Assurance Methodology

The verification process included interviews with NuLife staff responsible for collecting, monitoring, and reporting operational and carbon removal data. The audit team, led by Leonard Barkley (Lead Auditor), reviewed NuLife's data collection procedures, tracking systems, and methodologies applied under the Isometric program.

Verification activities included a review of feedstock documentation, HTL process data, sampling records, bio-oil transport documentation, storage and injection records, energy use data, and other supporting evidence. Both on-site observations and desk-based document reviews were performed.

3.3 Independent Statement

We confirm that Normec Verifavia and the verification team are independent of both NuLife and Isometric. Normec Verifavia has not participated in preparing feedstock data, carbon accounting calculations, monitoring records, or any documentation submitted to or from Isometric, other than this Verification Statement.

3.4 Assurance Opinion: Supporting Data & Monitoring Methodology

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The verification process concludes that NuLife's monitoring and data management methodology is fit for purpose and consistently applied. Reported data is supported by accurate, traceable, and reliable evidence, and appropriate quality assurance and quality control procedures are in place.

Period	Biomass slurry (mixed with water) Processed (Liters)	Bio-oil Produced (kg)	Carbon Sequestered (tCO_2e)	Process and project emissions (tCO2e)	Net Carbon Sequestered (tCO_2e) (table does not include project establishment and end-of-life emissions)
23 Oct 2025 to 30 Oct 2025	66,216	25,571	61.794	2.094	59.55
20 Oct 2025 to 22 Oct 2025	66,457	24,987	69.271	1.939	67.19
26 Sep 2025 to 19 Oct 2025	78,306	21,650	57.639	2.385	55.11
04 Sep 2025 to 25 Sep 2025	73,249	22,121	60.637	2.152	58.34

3.5 Verification Opinion

The objective of this verification was to confirm that NuLife's performance attributes and carbon removal claims are real, quantifiable, and verifiable, and that the project has met the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and its associated modules.

During the auditing period, all findings raised in the issues log were resolved. Normec Verifavia identified 3 clarifications, omissions, or missing requirements related to submitted documents and data. A detailed breakdown of these issues and their resolution is provided in Appendix 1.

In conclusion:

- NuLife has prepared and submitted its GHG Statement to Isometric free of material misstatement; and
- NuLife has prepared and submitted its GHG Statement in full conformance with the requirements of the Isometric Bio-Oil Geological Storage Protocol (V1.1) and associated modules.

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The result is a Positive Verification Statement, the basis for which is detailed in this verification report and the accompanying evidence.

Leonard Barkley

Lead Auditor

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Criteria	Evidence	Protocol / Standard Section	Compliant	Comments (unchanged since initial validation)
Scope	Project activity is HTL conversion of waste biomass into bio-oil, with permanent storage in [redacted] salt caverns [redacted] Eligible feedstocks include DAF sludge, oat hulls, bran, agricultural residues, FOG, etc. Boundaries cover Saskatoon HTL facility and injection operations.	Isometric Bio-Oil Geological Storage Protocol v1.1, Sections 1.2–1.3	Yes	Scope clearly defined for facility and storage site. Transport and ancillary emissions included.
Additionality	Project is financially additional: revenues from credits are required for commercial viability. Regulatory additionality confirmed: no laws mandate HTL or geological storage of bio-oil. Common practice additionality supported: few/no comparable operations in Canada. TRL internally assessed at 8; benchmarked against U.S. DOE (to be confirmed).	Standard Section 2.5.3	Yes	NuLife does not generate any revenue from the sale of bio-oil and is not financially viable without the generation and sale of carbon credits. There is no current mandate by the province or the country that enables to production of bio-oil from waste biomass

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Uncertainty	Monitoring plan includes Option B sampling: bio-oil sampled ~3x daily, duplicates tested. Carbon wt% verified by elemental analysis at University of Saskatchewan. Brine tested monthly; continuous flow meters for injection volumes.	Standard Section 2.5.8	Yes	Monitoring plan robust, but external lab not ISO 17025 accredited. NuLife uses a lab that has provided the necessary details Isometric requested for non-ISO 17025 labs.
Permanence	Salt caverns in Prairie Evaporite Formation; regulated under CSA Z341.2. Caverns pressurized at ~300 psi; mechanical integrity testing every 5 years. Storage security >1,000 years. Pressure monitoring continuous; no negative pressure events recorded.	Protocol Section 2.5.8	Yes	Permanence supported by geological evidence and regulatory oversight. Additional documentation on potash seams/leakage history recommended.
Leakage (market effects)	Feedstocks are local waste residues (e.g., oat hulls, dairy DAF sludge, grain milling residues, FOG and oat hulls) that are not bought/sold as commodities and have no alternative energy/materials market; project does not alter supply or demand elsewhere. Supplier confirmations of waste status and no double-claiming: [redacted]	Standard Section 2.5.4 Feedstock accounting module	Yes	No carbon market leakage expected because inputs are non-market wastes; crediting excludes any biomass that would remain durably stored or be used for energy. Physical risks are covered under

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	[redacted] documentation certifying that feedstocks are ineligible for energy use per protocol.			Permanence/Risk of Reversal, not Leakage.
Ownership	NuLife holds operational control of HTL plant. [redacted] owns and operates storage caverns (agreement in place April 2025). Letters/contracts needed from [redacted] confirming no other claims on environmental attributes.	Protocol Section 3.1	Yes	Ownership arrangements have been shared.

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Appendix

Appendix 1: List of Issues

#	Type	Date Issued	Issue Description	Resolution	Impact on material misstatement? (Y/N)	Impact on non-conformance? (Y/N)	Impact on adverse verification statement if not addressed? (Y/N)	Date Resolved
1	Clarification	12-Nov-2025	Electricty - Where does the value 6124 kWh come from for the energy consumption in the period Oct 17-29. consumption	The 395 kWh comes from the Oct 30th trial, where 37% of the batch was allocated to the injection. Oct 30th was the last trial included, and only 37% applies because all trials feed into a single oil tank, so exact matching isn't possible. There was also a typo in the original table: it showed *6,124 kWh × 0.37 = 395 kWh*. This has been corrected to *1,068 kWh × 0.37 = 395 kWh*.	Y	N	Y	12-Nov-2025
2	Clarification	17-Nov-2025	Biomass sourcing - Can you tell me	Feedstock transport events are included at 100% in the period they occur. For example, if an injection batch happens	N	N	N	17-Nov-2025

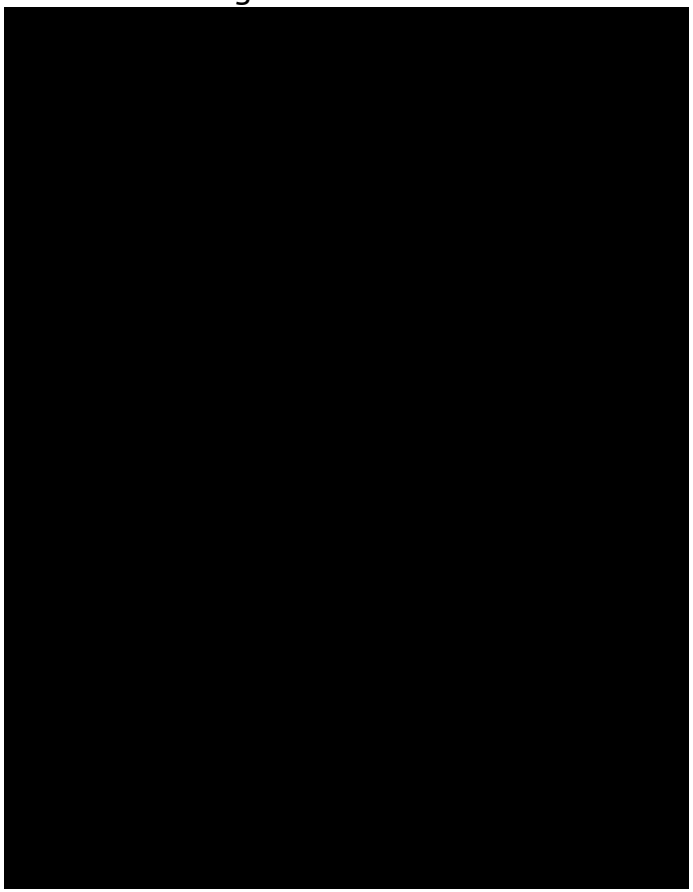
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			why there is only one event in Certify for the current GHG statements but two for the previous one ?	on November 30 and bio-oil was produced throughout November, we include the two feedstock deliveries from that month. If the next injection batch is on December 31 and bio-oil was produced throughout December, we include the one delivery from that month.				
3	Clarification	17-Nov-2025	Carbon sequestration results - When computing the values for each Batch, I find a slightly difference values than that of Certify: For Batch 005 I find: 22,121.43kg of Bio-oil * 74.75% carbon content * 3.67 = 60,682.2 kgCO2e While Certify claims 60,637 kgCO2e.	This might be because the formula is technically mass x carbon wt% x 3.667	Y	Y	Y	17-Nov-2025

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Appendix 2. Pictures from site visit

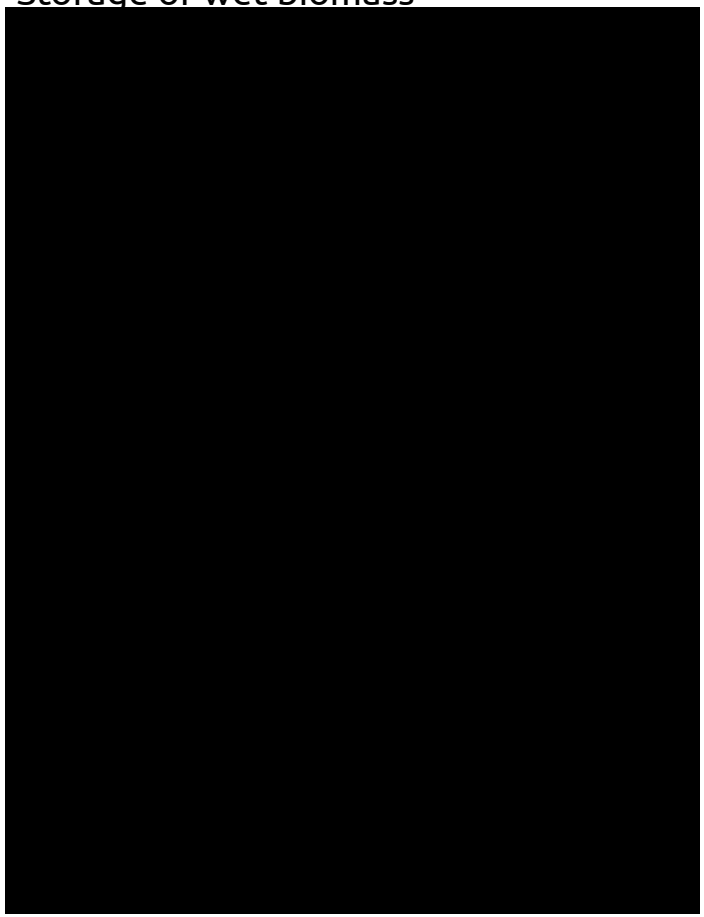
Storage tank of feedstock



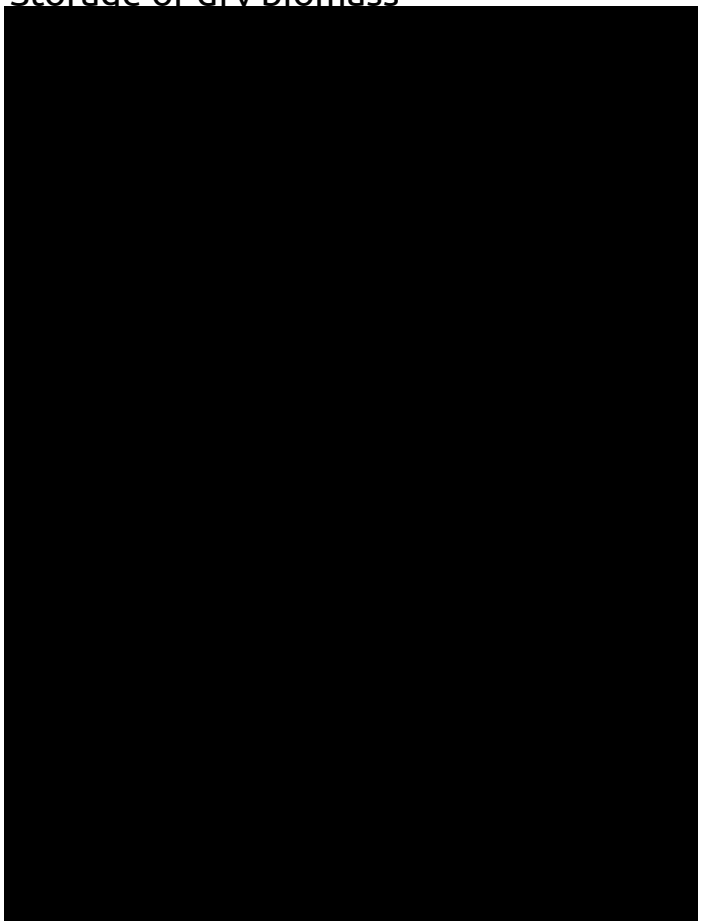
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Storage of wet biomass



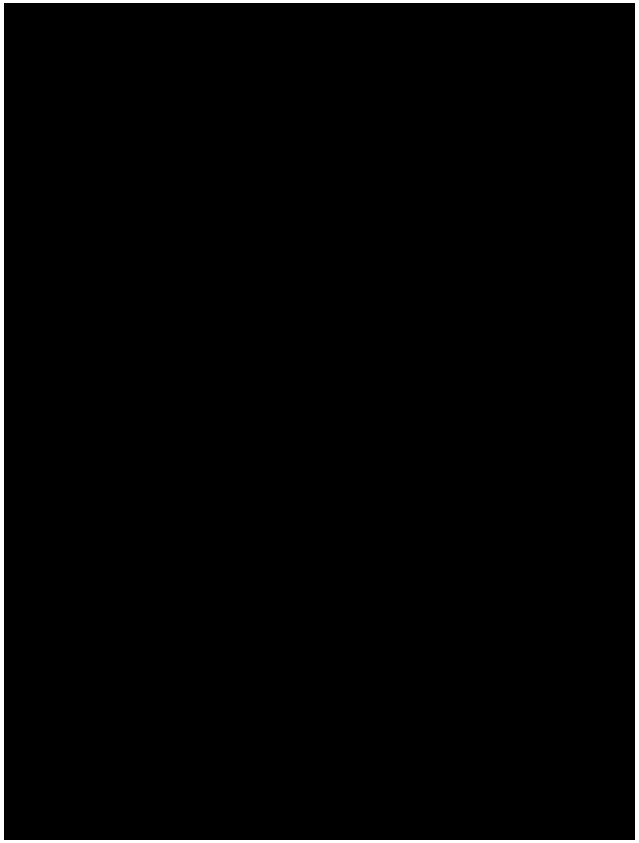
Storage of dry biomass



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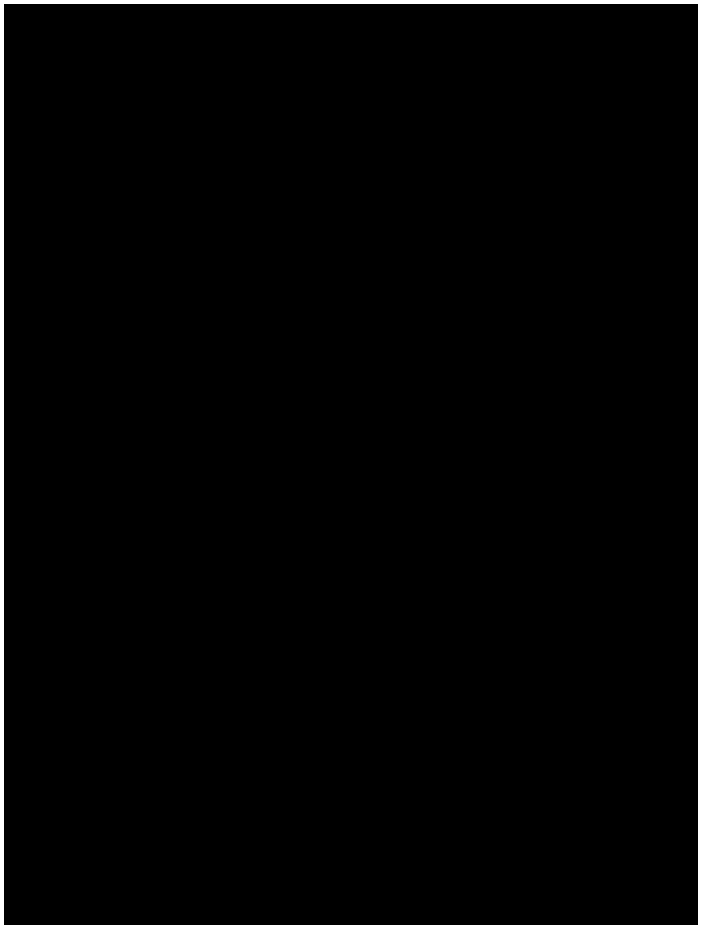
Hydrothermal liquefaction processors (liquid biomass flows back and forwards through the pipes)



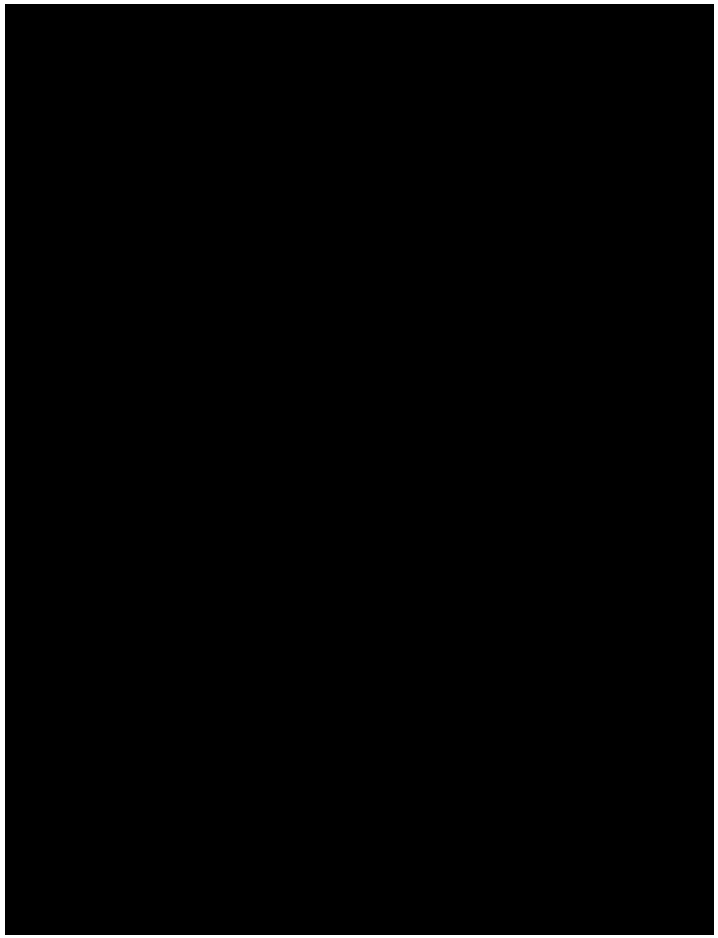
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Flow diagram of NuLife’s inputs and outputs.



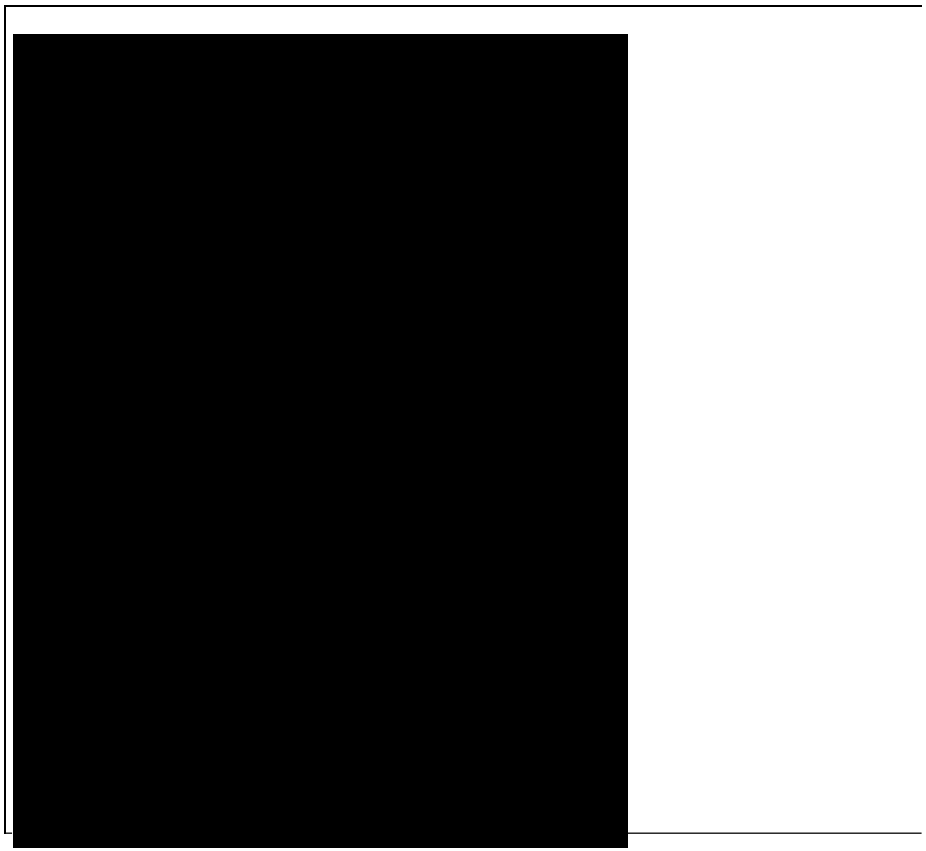
Example of tracking of cumulative energy consumed over given period.

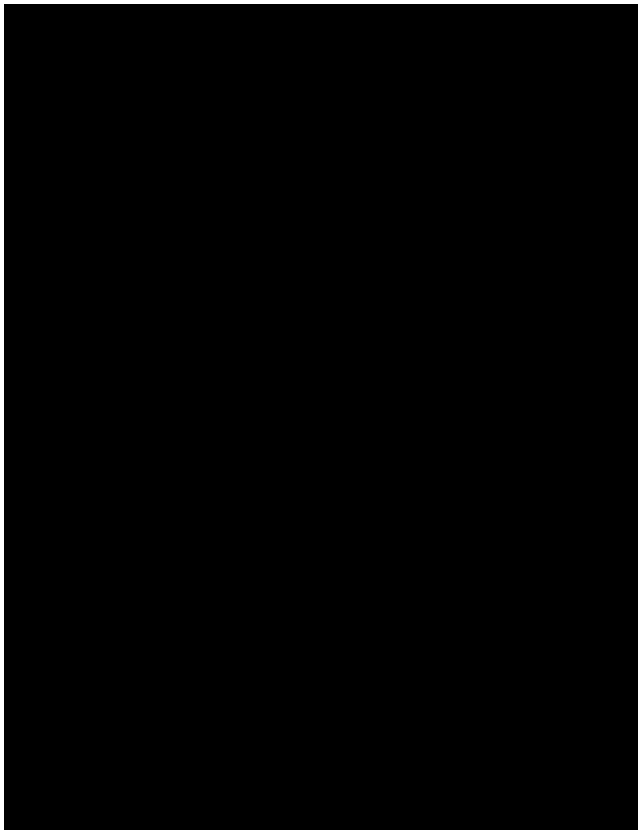


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Main power meter





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